# Malcolm Clottey

Software Engineer | Full-Stack, Cloud & Applied AI | AWS Certified Developer
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## Summary

I build software end to end, from client requirements to production. I led a team of 5 to build ClearLegacy, an AI tool on AWS that found all 4 planted issues in ~30 seconds at ~$0.05 per review, backed by 100+ automated tests. I've built OCR + RAG pipelines for Pfizer with 100% citation accuracy, 4 connected AI agents for Wayfair, a Next.js data platform serving 6,000+ pages at sub-150ms loads, and 5 Dockerized services deployed via CI/CD to AWS with 100% uptime. Computer Science graduate (Ashesi University, Major GPA 3.71, Dean's List), M.S. Computer Science student at Johns Hopkins starting Spring 2027, and AWS Certified Developer - Associate.

## Technical Skills

**Languages & Frameworks:** Python, Java, JavaScript/TypeScript, SQL, C++, React, Next.js, Node.js/Express, Django, Flask, FastAPI, REST APIs
**Cloud, Data & AI:** AWS (Bedrock, Lambda, S3, DynamoDB, Textract, EC2), Docker, CI/CD, PostgreSQL, MySQL, LLMs, RAG, AI agents (n8n), ML (PyTorch, XGBoost)
**Practices:** Automated testing, monitoring (Prometheus, Grafana), security (encryption, PII masking, JWT auth), Git, Agile/Scrum, stakeholder communication

## Experience

**AI Document Intelligence Extern (in partnership with Pfizer)** Jun 2026 - Present
*Extern, Inc.* *Remote*

- Built AI pipelines using OCR and RAG to process and classify pharmaceutical vendor documents, achieving 100% Recall@K and 100% Hit Rate in retrieval evaluation (n=5) with a 0.840 Mean Reciprocal Rank and 100% citation accuracy.
- Benchmarked end-to-end answer accuracy at 83% against manually-verified ground truth and profiled pipeline latency to isolate LLM inference as 99.98% of total response time, delivering a technical report on trade-offs and deployment recommendations.

**AI Extern (in partnership with Wayfair)** Mar 2026 - May 2026
*Extern, Inc.* *Virginia, United States*

- Built and operated 4 connected AI agents in n8n (trend discovery, competitor monitoring, AI content generation, and an integrated dashboard) automating market intelligence for Wayfair's Rugs category team.
- Surfaced a near-total Amazon whitespace opportunity (0 in-category products analyzed) and mapped competitor pricing tiers, then built a live-updating Google Sheets dashboard (React + Apps Script) to surface the findings.

**Software Developer Intern** Jul 2025 - Sep 2025
*Tyncad Software Company*

- Engineered full-stack improvements on a live web platform using Django and modern JavaScript, enhancing system scalability and maintainability.
- Collaborated cross-functionally to integrate redesigned UI with dynamic back-end endpoints, ensuring seamless data flow and clean separation of concerns.
- Refactored front-end assets and minimized HTTP payloads, slashing load times by approximately 40%.

## Projects

**ClearLegacy - LPL Financial University Hackathon** Oct 2026
*Project Lead*

- Led a team of 5 to build an AI tool that compares clients' estate documents with account records and flags beneficiary conflicts for advisor review, finding all 4 planted issues in ~30 seconds at ~$0.05 per review.
- Built the Amazon Bedrock (Claude) extraction layer with a quote-verification step that discards unsupported facts, plus Textract OCR, encrypted S3, a DynamoDB audit log and Comprehend PII masking, backed by 100+ automated tests.

**Global Remote Compliance & Salary Index** Jul 2026 - Present
*Full-Stack Engineer*

- Designed and shipped a B2B data platform end to end, generating 6,000+ programmatic pages via Next.js Static Site Generation with sub-150ms page loads.
- Built a Node.js/Python data pipeline integrating public APIs into structured, auditable database seeds.

**Hybrid Algorithmic & AI-Based Timetabling System** Jan 2025 - Apr 2025
*Project Lead*

- Designed a scheduling engine (greedy + genetic algorithms) assigning 483 sessions across 20 rooms in under 30 seconds.
- Deployed 5 Dockerized services via CI/CD to AWS EC2 (Linux) with 100% uptime, monitored with Prometheus/Grafana.

## Education

**M.S., Computer Science - Engineering for Professionals** Expected December 2028
*Johns Hopkins University* *Baltimore, MD*

**BSc., Computer Science** Jan 2022 - Jun 2025
*Ashesi University College - Major GPA: 3.71/4.00 - Dean's List*

## Certifications

**AWS Certified Developer - Associate** | Amazon Web Services | Sep 2025 - Sep 2028